

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	Total Marks:	100
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
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Section / Batch:	Btech 12024	Date:	12/08/26

Code of Conduct

I Dharmika / 192421094 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Dharmika

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Terrance Frederick Fernandez	Terrance Frederick Fernandez	
Signature: <u>[Signature]</u>	Signature: <u>[Signature]</u>	Signature: _____
Date: <u>12/8/26</u>	Date: <u>12/8/26</u>	Date: _____

95
100

Soln

Data packet size = 500 bytes

Tunnel overhead = 20 bytes

$$\left. \begin{array}{l} \text{total transmitted packet} \\ \text{size} \end{array} \right\} = 500 + 20 \\ = 520$$

$$\frac{20}{500} \times 100 = 4\%$$

Additional overhead : 20 bytes.

Soln

Bandwidth = 20 MHz

64-QAM = 6 bits / symbol

$$\begin{aligned} \text{Data rate} &= 20 \text{ MHz} \times 6 \text{ bits / symbol} \\ &= 20 \times 10^6 \times 6 \end{aligned}$$

$$= 120 \times 10^6$$

1 bit - 2-QAM
2 bit - 4-QAM
6 bit - 64-QAM

Maximum Theoretical data rate = 120 Mbps

$\therefore$ 64-QAM = 64-Quadrature Amplitude Modulation

uses

WiFi, Digital TV, 4G/5G networks,
cable internet

2)

Soln

Number of Entries = 50

Memory per Entry = 32 bytes

$$\begin{aligned}\text{Total memory} &= 50 \times 32 \\ &= 1600 \text{ bytes}\end{aligned}$$

Convert to KB

$$1600 \div 1024 \approx 1.56 \text{ KB}$$

Proactive Routing

A routing protocol that maintains route continuously, Even before they are needed

Ex: OLSR - Optimized Link State Routing Protocol

Reactive Routing

A routing protocol that finds a route only when it is needed

Ex: ASR - Dynamic Source Routing

Soln

$$\text{Speed} = 10 \text{ m/s}$$

$$\text{update interval} = 2 \text{ seconds}$$

Formula

$$\begin{aligned}\text{Distance} &= \text{Speed} \times \text{Time} \\ &= 10 \times 2 \\ &= 20 \text{ m}\end{aligned}$$

$\therefore$ 20 meters between routing updates.

$$\text{Total RREQ packets} = 5 + 1 \\ = 6$$

$$\therefore \text{Route Request packet} = 6$$

In DSR it is forwarded by each intermediate node

Soln

$$\text{Subset mask : } 255 \cdot 255 \cdot 255 \cdot 240$$

Convert 240 to binary

$$240 = 11110000$$

$$\therefore \text{No of host bits} = 4$$

$$\text{formula : } 2^{\text{host bits}} = 2^4 = 16$$

$$16 - 2 = 14$$

Soln

$$\text{Spectral Efficiency} = \frac{\text{Data Rate}}{\text{Bandwidth}}$$

$$= \frac{1}{20}$$

$$= 0.05 \text{ bits}$$

MSPS $\rightarrow$ Mega symbol per seconds

8)

soln

Number of nodes = 50

Area = 1 km^2

$$\text{Density} = \frac{\text{Number of nodes}}{\text{Area}}$$

$$= \frac{50}{1} = 50$$

Network density = 50 nodes/km^2

using node

$$r = 250 \text{ m}, \text{ Area} = 1 \text{ km}^2 = 10^6 \text{ m}^2$$

$$= \frac{50 \times \pi \times (250)^2}{10^6} = \frac{50 \times 3.1416 \times 62500}{10^6} \approx \frac{9817500}{10^6}$$

$$\approx 9.82$$

$\therefore 9.82$ (nodes per transmission area)

Soln

Mobile node speed,

$$v = 30 \text{ m/s}$$

Frequency

$$f = 900 \text{ MHz}$$

$$= 900 \times 10^6 \text{ Hz}$$

Speed of Light

$$c = 3 \times 10^8 \text{ m/s}$$

Doppler shift

$$f_d = \frac{vf}{c}$$

$$f_d = \frac{30 \times 900 \times 10^6}{3 \times 10^8}$$

$$f_d = 90 \times 10^2$$

$$f_d = 9000 \text{ Hz}$$

$$f_d = 9 \text{ KHz}$$

Doppler shift is the change in the frequency of a received signal caused by the movement of the transmitter, receiver or both

10)

Soln

Data rate = 54 Mbps

Packet size = 1500 bytes

Convert bytes to bits

$$1500 \times 8 = 12000 \text{ bits}$$

Transmission time

$$T = \frac{\text{packet size (bits)}}{\text{Data rate (bits)}}$$

$$= \frac{12000}{54 \times 10^6}$$

$$T = 0.000222 \text{ s}$$

Convert to microseconds

$$0.000222 \times 10^6 = 222.2 \text{ ns}$$

$$\text{Transmission time} = \underline{222.2 \text{ microseconds (ns)}}$$